Alcohol & Health Outcomes

Statistical learning with BRFSS and U.S. Chronic Disease Indicators

James Riggs | Team course project; teammate analysis preserved as Jose.qmd

01 Research question

How do demographic, socioeconomic, and health characteristics relate to binge drinking? Separately, how do state-level health and income patterns relate to alcohol-related outcomes?

02 Data & scope

50,000

BRFSS input survey rows, before cleaning.

USCDI: state-level alcohol/liver indicators. The project summary describes approximately 21,500 analysis records.

The saved USCDI extract has 50,000 rows; the exact filtering stage for the reported 21,500-record scope is unverified. Neither count is a final model sample size.

Individual survey responses and aggregate state indicators are separate units of analysis.

03 Methods

R / tidymodels / Quarto
Regularized logistic regression (glmnet) and teammate random-forest work (ranger). Sources specify five-fold cross-validation and ROC AUC tuning.

Prepare
Recode survey responses and prepare predictors

Model
Fit and tune classification workflows

Compare
Explore individual and state-level associations

State-level workflow: aggregate BRFSS measures by state, then join to USCDI indicators.

04 Reported findings

Younger adults
The project reported higher odds of binge drinking among younger adults.

State-level patterns
The project reported associations between poorer general health, lower income, and alcohol-related deaths.

Qualitative findings are from the supplied project summary. Final effect sizes, uncertainty intervals, and held-out metrics were unavailable for verification.

05 Presentation & limits

Presented by James Riggs to Cal Poly Statistics faculty. The work connected data preparation, statistical learning, and communicating limitations.

Observational associations do not establish causation. State-level patterns do not establish individual risk. Chronic liver disease mortality is broader than alcohol-attributable mortality.

Saved sources require coding and evaluation review, including missing-value handling, outcome labels, an undefined recipe, and target leakage in one state-level workflow.

Sources: Classification_project(James).qmd; Jose.qmd; project summary supplied by James Riggs. Source audit and reproduction notes are in the project repository.

Case study, sources & poster
jriggs03.github.io/stat551.html